

MANUFACTURER TEST CERTIFICATE

(According to DIN EN 10204 TYPE 3.1- 2004)

Certificate No : CDI/LAB/2858C Date: 07.11.2023	W.O. No : 2858
Client: M/s. DODSON GLOBAL INC	Inspection Agency : CDI
PO No. 362947-408 DATE 06.09.2023	Order Status : P
Material Specification: ASTM A105 EDITION - 2021 / SA105OF ASME BPVC SEC-II, PART-A. EDITION-2021	

We hereby certify that the material detailed below meets the requirement of purchase order/relevant specification

S. No	Item Description	Material	Offer Qty.	Batch/Lot/Group No.	Heat Treatment type	H.T. Soaking Temperature	H.T. Soaking Time	Cooling Medium
16	4" - 900# - WNRF - SCH XS	SA/A105N	75	A2975A	Normalized	910°C	2.00 Hrs.	Air
19	3" - 1500# - WNRF - SCH XS	SA/A105N	225	A2976A	Normalized	910°C	2.00 Hrs.	Air
29	30" - 150# - BLRF - B16.47 SER-B	SA/A105	19	N2106	As forged	-	-	-
29	30" - 150# - BLRF - B16.47 SER-B	SA/A105	1	N2107	As forged	-	-	-
39	10" - 1500# - BLRF	SA/A105N	2	A2978A	Normalized	910°C	6.00 Hrs.	Air
39	10" - 1500# - BLRF	SA/A105N	3	A2978B	Normalized	910°C	6.00 Hrs.	Air
49	12" - 1500# - WNRTJ - SCH 120	SA/A105N	2	A2978C	Normalized	910°C	6.00 Hrs.	Air
49	12" - 1500# - WNRTJ - SCH 120	SA/A105N	2	A2978D	Normalized	910°C	6.00 Hrs.	Air
49	12" - 1500# - WNRTJ - SCH 120	SA/A105N	2	A2979A	Normalized	910°C	6.00 Hrs.	Air
49	12" - 1500# - WNRTJ - SCH 120	SA/A105N	2	A2979B	Normalized	910°C	6.00 Hrs.	Air
49	12" - 1500# - WNRTJ - SCH 120	SA/A105N	2	A2979C	Normalized	910°C	6.00 Hrs.	Air
50	2" - 2500# - WNRTJ - SCH 160	SA/A105N	295	A2977A	Normalized	910°C	2.30 Hrs.	Air
		TOTAL	630	Nos				

CHEMICAL COMPOSITION (%)

Batch/Lot/ Group No.	Heat/Cast/ Melt No.	Lab Ref No.	C	Mn	P	S	Si	Cr	Ni	Mo	V	Cu	AL	CE**
					LADLE ANALYSIS									
	23E00832	15362	0.200	1.080	0.020	0.007	0.280	0.005	0.001	0.001	0.001	0.002	0.050	0.38
	G4/23	15405	0.200	0.980	0.021	0.009	0.280	0.260	0.040	0.020	0.005	0.030	0.022	0.43
	23E01177	15346	0.210	1.130	0.018	0.007	0.21	0.006	0.001	0.001	0.001	0.001	0.023	0.40
	CHP22648	15294	0.210	0.730	0.021	0.013	0.23	0.210	0.090	0.033	0.009	0.032	0.032	0.39
	CHP23226	15409	0.210	0.780	0.032	0.025	0.27	0.180	0.044	0.020	0.003	0.036	0.048	0.39
	CHP23233	15408	0.190	0.750	0.024	0.016	0.25	0.150	0.060	0.020	0.008	0.040	0.032	0.36
	J-331	15422	0.193	0.862	0.025	0.016	0.215	0.184	0.024	0.007	0.000	0.031	0.023	0.38
	CHP23162	15386	0.210	0.760	0.025	0.014	0.240	0.200	0.060	0.020	0.005	0.040	0.032	0.39
	CHP23169	15385	0.195	0.790	0.028	0.019	0.310	0.190	0.040	0.040	0.005	0.040	0.037	0.38
	CHP23233	15394	0.190	0.750	0.024	0.016	0.250	0.150	0.060	0.020	0.008	0.040	0.032	0.36
	H1/21	15419	0.180	0.950	0.020	0.009	0.230	0.280	0.030	0.020	0.006	0.020	0.022	0.40
Specification Requirement		Minimum	-	0.60	-	-	0.10	-	-	-	-	-	0.020	-
		Maximum	0.35	1.05*	0.035	0.040	0.35	0.30	0.40	0.12	0.08	0.40	-	0.43



**CHEMICAL COMPOSITION (%)**

Batch/Lot/ Group No.	Heat/Cast/ Melt No.	Lab Ref No.	C	Mn	P	S	Si	Cr	Ni	Mo	V	Cu	AL	CE**
PRODUCT ANALYSIS														
A2975A	23E00832	15362	0.207	0.970	0.016	0.002	0.234	0.175	0.025	0.022	0.005	0.020	0.026	0.41
A2976A	G4/23	15405	0.201	0.990	0.021	0.009	0.275	0.258	0.041	0.024	0.014	0.020	0.020	0.43
A2977A	H1/21	15419	0.195	0.972	0.018	0.009	0.269	0.255	0.040	0.024	0.014	0.020	0.020	0.42
A2978A	CHP23226	15409	0.230	0.698	0.008	0.002	0.349	0.202	0.245	0.023	0.078	0.072	0.020	0.43
A2978B	CHP23233	15408	0.218	0.994	0.005	0.002	0.319	0.188	0.098	0.022	0.004	0.027	0.020	0.43
A2978C	J-331	15422	0.162	0.816	0.022	0.014	0.212	0.186	0.026	0.022	0.005	0.035	0.023	0.34
A2978D	CHP23226	15409	0.230	0.698	0.008	0.002	0.349	0.202	0.345	0.023	0.078	0.072	0.020	0.43
A2979A	CHP23162	15386	0.221	0.760	0.020	0.012	0.244	0.262	0.108	0.034	0.009	0.035	0.021	0.42
A2979B	CHP23169	15385	0.181	0.797	0.024	0.015	0.270	0.252	0.027	0.032	0.005	0.031	0.030	0.38
A2979C	CHP23233	15394	0.228	0.765	0.018	0.007	0.274	0.152	0.043	0.022	0.005	0.027	0.031	0.40
N2106	23E01177	15346	0.215	1.14	0.015	0.002	0.212	0.090	0.019	0.022	0.005	0.020	0.030	0.43
N2107	CHP22648	15294	0.215	0.714	0.021	0.012	0.230	0.109	0.047	0.022	0.005	0.032	0.021	0.37
Specification Requirement	Minimum	-	0.60	-	-	-	0.10	-	-	-	-	-	0.020	-
	Maximum	0.35	1.05*	0.035	0.040	0.35	0.30	0.40	0.12	0.08	0.40	-	0.43	

PHYSICAL PROPERTIES

Batch/Lot/ Group No.	Yield Strength (MPa) (0.2% offset)	Tensile Strength (MPa)	Elongation % on GL=4D	Reduction in Area %	Hardness (B.H.N.)	Test Sample Direction	Charpy V- Notch – 10x10x55mm (2mm V- Notch)				Remark
							I	II	III	Average (In Joules)	
A2975A	319	515	31.00	59.93	149, 153	Tangential	-	-	-	-	Satisfactory
A2976A	342	538	31.00	58.01	153, 156	Tangential	-	-	-	-	Satisfactory
A2977A	341	520	30.40	58.14	153, 156	Tangential	-	-	-	-	Satisfactory
A2978A	350	538	31.00	59.04	156, 159	Tangential	-	-	-	-	Satisfactory
A2978B	326	521	31.40	56.97	153, 156	Tangential	-	-	-	-	Satisfactory
A2978C	320	517	31.20	57.67	149, 153	Tangential	-	-	-	-	Satisfactory
A2978D	334	529	31.20	59.04	153, 156	Tangential	-	-	-	-	Satisfactory
A2979A	342	521	30.40	58.01	153, 156	Tangential	-	-	-	-	Satisfactory
A2979B	335	531	31.00	58.91	156, 159	Tangential	-	-	-	-	Satisfactory
A2979C	350	538	31.00	59.04	156, 159	Tangential	-	-	-	-	Satisfactory
N2106	342	538	30.20	58.01	156, 159	Tangential	-	-	-	-	Satisfactory
N2107	319	523	30.00	58.91	153, 156	Tangential	-	-	-	-	Satisfactory
Specification Requirement	250	485	22	30	137	-	-	-	-	-	Min Req
	-	-	-	-	187	-	-	-	-	-	Max. Req.



**Remarks: -**

- 1- All Material are as per ASTM A105 EDITION - 2021 / SA105 OF ASME BPVC SEC-II, PART-A EDITION-2021.
- 2- Visual checked and found free from Cracks, sharp Notches and dent and Mechanical Marks.
- 3- Dimension 100% checked and found acceptable limit as per ASME B16.5 Edition 2020 & ASME B16.47 SER-B Edition 2020.
- 4- Material is fully killed & fine grain practice.
- 5- Welding has not been performed on Flanges.
- 6- All items meeting general requirement of ASTM 961/ASTM A961M EDITION – 2021 / SA961/SA961M EDITION–2021
- 7- Flange Gasket Finishing on RF Flanges 125 μ m (3.2 μ m) -250 μ m (6.3 μ m) and RTJ Flanges 63 μ m (1.6 μ m).
- 8- Job and Groove Hardness checked 2 Nos minimum each Size, per item, per heat treatment lot and found 149 to 165 BHN.
- 9- Material is confirms to NACE MR 0175 /ISO 15156-2015 and NACE MR 0103/ ISO 17945-1:2015.
- 10- All Requirements of material specification and CSA Z245.12-21 Grade 248 Cat-I have been fully complied with.
- 11- *C.E. Formula = $C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$.
- 12- *For each reduction of 0.01% below the specified carbon maximum (0.35%), an increase of 0.06% manganese above the specified Maximum (1.05%) will be Permitted up to 1.35 % For SA105 and 1.65 % For ASTM A105.
- 13- Mechanical Test Sample Preparation as per SA370/ASTM A370EDITION-2021.
- 14- Melting of origin – INDIA.
- 15- MARKING As per MSS SP-25-2018 :

16	CDI	4" - 900# SCH XS	SA/A105N	HEAT CODE-A2975A	B16.5	NACE	INDIA
19	CDI	3" - 1500# SCH XS	SA/A105N	HEAT CODE-A2976A	B16.5	NACE	INDIA
29	CDI	30" - 150#	SA/A105	HEAT CODE-N2106	B16.47 SER-B	NACE	INDIA
29	CDI	30" - 150#	SA/A105	HEAT CODE-N2107	B16.47 SER-B	NACE	INDIA
39	CDI	10" - 1500#	SA/A105N	HEAT CODE-A2978A	B16.5	NACE	INDIA
39	CDI	10" - 1500#	SA/A105N	HEAT CODE-A2978B	B16.5	NACE	INDIA
49	CDI	12" - 1500# SCH 120	SA/A105N	HEAT CODE-A2978C	B16.5	NACE	INDIA
49	CDI	12" - 1500# SCH 120	SA/A105N	HEAT CODE-A2978D	B16.5	NACE	INDIA
49	CDI	12" - 1500# SCH 120	SA/A105N	HEAT CODE-A2979A	B16.5	NACE	INDIA
49	CDI	12" - 1500# SCH 120	SA/A105N	HEAT CODE-A2979B	B16.5	NACE	INDIA
49	CDI	12" - 1500# SCH 120	SA/A105N	HEAT CODE-A2979C	B16.5	NACE	INDIA
50	CDI	2" - 2500# SCH 160	SA/A105N	HEAT CODE-A2977A	B16.5	NACE	INDIA



Signature /Seal Of Inspector
(For-Inspection Agency)

Manager (Q.C)
(C.D. Industries)